



A sporty humanoid—more than a toy, really!

point of view. Of course, for toys, the aesthetic value is high too," says T. Jayakrishnan, director and country manager, Energid Robotics and Machine Vision.

Although this trend has caught on in India only in recent years, the history of robotic toys dates back to many years ago. "The first toy robot is believed to be the yellow, boxy, clockwork called Robot Lilliput, invented and mass-manufactured in Japan sometime between 1930 and 1940. It was followed by another Japanese toy called the Atomic Robot Man," says Jayakrishnan. Replicas of these are still available in certain foreign markets and online stores.

Of course, today's robotic and computerised toys are much more complex. They seem more like a set-up of numerous electronic and mechanical components, often even microprocessor boards, in the guise of a happy toy! Most of these toys are not just entertaining but educational as well, since they boggle the child's

brain and foster new ideas and technical skills.

Are these toys or robots?

Look at some of the high-tech toys available in the market today and you really cannot differentiate between a toy and a robot. One of the best examples of the current-generation high-tech toys is Lego Mindstorm's NXT 2.0 Intelligent Brick, which has four sensor inputs and three actuator outputs. It is a robot toy you can build and program. The programming is done through a graphical user interface, which lets you fit together logic building blocks like a flowchart. Once the flowchart is ready, the code can be downloaded into the machine to perform the task. The program can be modified and dumped into the machine any number of times till the desired output is achieved.

WowWee's Robopanda is another example of an entertaining robot toy. It is a playful and talkative interactive friend. Robopanda loves to share stories and jokes, play games, sing songs and talk with children of all ages. It can even crawl on all fours and return to a sitting position. Its working is controlled by interactive cartridge-based content. The user can expand the repertoire of Robopanda's stories, games and activities by changing activity and story cartridges (sold separately). WowWee has other toys such as Robosapien, Dragonfly and Roboraptor.

"Tomy's iSobot and Sony's Rolly are other good examples. Not to forget Roboraptor and Pleo—the dinosaur toys," says Jayakrishnan. "WowWee's Roboraptor was launched in 2005. The 81.28cm (32-inch) toy has a multi-function remote control that utilises infrared (IR) technology. It is equipped with several sensors that let it interact with its surroundings. It can walk, move around obstacles and run. It can even collect items while running. In 2006, a more complex robotic

dinosaur toy called Pleo was launched by Innvo Labs. The toy imitates the appearance and behaviour of a young Camarasaurus. The toy's large skull conceals the many sensors and motors used for its movements. Pleo's artificial intelligence enables each toy to learn from its surroundings and acquire a unique personality. Jayakrishnan adds that Sony's AIBO, although discontinued now, was a wonderful toy or pet robot with emotions and impulses due to artificial intelligence.

"Outdoor toys like aero-modelling kits are often priced above Rs 30,000. These models have engines or high-performance brushless DC motors, and are controlled by high-end four-to fourteen-channel remote controls. The models might be exact replicas of fighter jets with real mini-sized jet engines on them. Cost of such toys could be around Rs 200,000," explains Fahad Azad, managing partner, Robosoft Systems. His team is currently working on a do-it-yourself robotic platform with which a child can make multiple kits out of a single set of components. The robot can be controlled through a wireless remote that connects various motors on the controller brain. This



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